DEPENDABILITY OF INFORMATION 12

In chemistry you need to be able to:

- design and carry out first-hand investigations
- discuss first-hand investigations
- collect information from secondary sources
- assess information from secondary sources
- decide on the reliability of the results you obtain from your work
- decide on the reliability of any information your results suggest.

First-hand investigations are those that you carry out yourself, e.g. practicals in the laboratory. Secondary sources include textbooks and encyclopaedias.

1 Name two other sources of secondary information.

2 Many organisations do not allow people to begin scientific investigations until they have checked existing (secondary) information. Explain why.

3 Name two sources of secondary information you could use for designing and carrying out investigations on each topic.

a Which fertiliser works best on a particular crop?

b Which catalyst best speeds up a particular reaction?

4 Many people rely on recommendation or 'word of mouth' for information. How do these sources differ from other secondary sources of information?

Variables

During experiments, we often concentrate on observing and recording what happens when conditions change. Factors that can be changed during an experiment are called **variables**. Usually only one variable is changed at a time during an experiment. There are four major groups of variables.

- independent
- dependent
- controlled
- uncontrolled.

The **independent** variable is the major part of the hypothesis (belief) that you are testing. It is the condition that you change to see what happens. The **dependent** variable is the feature that changes as a direct result of changing the independent variable. A **controlled** variable is one that is kept the same in each experiment. An **uncontrolled** variable is one that has not been taken into account during the planning of the experiment.

5 A student noted that when zinc metal was added to dilute hydrochloric acid, hydrogen gas was given off. She observed that the smaller lumps of the metal seemed to react quicker than the larger lumps. She added 50.0 mL of 1.0 mol L^{-1} of hydrochloric acid to 5.00 g of zinc metal and measured how long it took for a volume of gas to be given off. She then repeated the experiment using the same mass of zinc but this time it was in powder form.

	Variables		
Hypothesis	Independent	Dependent	Controlled
Smaller particles react quicker than larger lumps	Size of zinc	Time for reaction to occur	Volume of acid Concentration of acid Mass of zinc

Table 1

a Study table 1, which summarises the variables in this experiment.

b Identify an uncontrolled variable in the experiment.

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6 A student wanted to test if increasing the concentration of acid increased the rate of evolution of gas in the reaction between zinc and hydrochloric acid. She added 5.00 g of zinc metal to 50 mL of acid and timed how long it took to evolve a fixed mass of gas. She repeated this, using different concentrations of acid. Her results are shown in table 2.
Using the format shown in table 1, identify the hypothesis, independent, dependent and controlled variables for this experiment.

Time (sec)	75	30	20	15	12
Concentration (mol L^{-1})	0.1	0.5	1.0	1.5	2.0

Table 2

Three terms are very important when it comes to assessing data:

- accuracy
- reliability
- validity

Accuracy: does not contain any errors
The mechanic was correct, he made an accurate *diagnosis of the fault in the car.*

Reliability: dependability; giving the same result every time the test is carried out
The indicator was reliable *– every time it was added to an acid it changed from pink to colourless.*

Validity: how relevant and appropriate a process is
She was convinced of the validity *of her method; it did measure what it was supposed to measure.*

7 Give another definition of *accuracy*.

8 Write a sentence illustrating the meaning of the word *accuracy* in a scientific context.

9 Discuss this statement: *A stopped clock is more accurate than one that is always exactly five minutes late.*

10 Sometimes results can be accurate to ± 5% or ± 10%.

a Give an example of when this level of accuracy (or slight inaccuracy) would be acceptable.

b Give an example of when 100% accuracy is required.

11 Give another definition of *reliability*.

12 Write another sentence illustrating the meaning of the word *reliability* in a scientific context.

13 Discuss this statement: *He set out for the bus, knowing he could rely on it being 5 minutes late.*

In science, reliability is discussed in terms of anyone using a particular method or following the same set of instructions getting the same results. These results might be incorrect if the method is flawed or the apparatus is not designed for that experiment. Data collection methods must be valid.

14 Give another definition of *validity*.

15 Write another sentence illustrating the meaning of the word *validity* in a scientific context.

16 Discuss this statement: *When he tried to board the plane they questioned the validity of his passport.*

17 A group of scientists declared that their investigations were reliable and that the conclusion they drew about a given drug trial was valid. A second group of scientists disputed the statement, claiming that the conclusions drawn were unreliable and therefore the conclusion was invalid. Given that both groups of scientists were competent, suggest why this type of dispute could occur.

18 Mark each statement as true or false.

a Dependent variables can change as a direct result of altering independent variables.

b A valid result is based on well-supported facts.

c All scientific results are valid if they are accurate to ± 5% or ± 10%.

d Controlled variables remain the same in each experiment.

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